Statistical Mechanics of the Rope Medium

Partition Function, Free Energy, Entropy, and the Finite-Temperature Defect Gas on One Functional

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Abstract

Thermodynamics was the continuum chain's last incomplete foundation: the corpus had a continuum energy, defects, and a phase transition named, but not the statistical mechanics that connects them. This paper supplies it, on the SAME functional $(K/2)\int|\nabla\theta|^2$ used throughout the chain. Because the coarse-grained orientation field has the same equilibrium energy functional as the two-dimensional XY model, its equilibrium statistical mechanics coincides with that of the XY model, which splits cleanly into two computable sectors. The Gaussian (spin-wave) sector gives the partition function as a Gaussian integral, from which the free energy per site, the entropy $S = -\partial F/\partial T$, the internal energy $U = F + TS$ (matching equipartition to machine precision), and the spin-wave specific heat all follow. The defect (vortex) sector gives the finite-temperature physics: the Berezinskii–Kosterlitz–Thouless transition at $T_{BKT} = \pi K/2$, the algebraic correlation exponent $\eta = T/2\pi K$ reaching the universal value 1/4 at the transition, the universal helicity-modulus jump $K_R(T_{BKT}) = 2T_{BKT}/\pi$ (self-consistently equal to K), and the Kosterlitz renormalization-group flow that separates a low-temperature phase of bound defect pairs from a high-temperature phase of free defects. All seven results are reproducibly computed (benchmarks/micromech/statistical_mechanics.py, 7/7). The transition is precisely the defect-gas unbinding of the defect-theory paper, so this completes the EQUILIBRIUM statistical mechanics and simultaneously ties together the defect, electromagnetic, and condensed-matter sectors that rest on it. The treatment is classical equilibrium statistical mechanics throughout; it does not touch the quantum boundary.

> **Scope of this paper**
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> CLAIM: on the same $(K/2)\int|\nabla\theta|^2$ functional, the rope medium's equilibrium statistical mechanics is computable in full — Gaussian partition function, free energy, entropy, energy, specific heat; and the BKT defect-gas transition with its universal exponent $\eta=1/4$, helicity jump, and RG flow. All seven reproducibly computed.
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> NOT CLAIMED: this is EQUILIBRIUM statistical mechanics only — nonequilibrium thermodynamics (transport, thermal conductivity, relaxation, fluctuation–dissipation, dynamics) is outside scope. Nothing quantum. The full non-perturbative XY partition function beyond the Gaussian + defect-gas decomposition is the established XY-model result, used and cited rather than re-derived.

1. The Missing Foundation

The continuum chain — microscopic mechanics, homogenization, effective field theory, electromagnetism, optics, defects — rested throughout on the energy functional $(K/2)\int|\nabla\theta|^2$, but treated it almost entirely at zero temperature: as an energy to be minimised, not a Boltzmann weight to be summed. That left thermodynamics as the one foundational sector supported by assertion rather than computation, even though it underpins the finite-temperature behaviour of the electromagnetic, condensed-matter, and defect sectors. This paper completes it.

The organising fact. The EQUILIBRIUM statistical mechanics of the coarse-grained orientation field coincides with that of the two-dimensional XY model, because the two share the same equilibrium energy functional. This is an equivalence of effective equilibrium theories, derived in Section 2a, not an analogy imported from outside and not a claim that the microscopic medium is a lattice of XY spins. Consequently the subject decomposes, exactly as the XY model does, into a Gaussian spin-wave sector (this section and the next) and a defect sector responsible for the finite-temperature phase transition (Sections 4–6). Everything below is computed on that one functional.

2. The Partition Function: Gaussian Sector

At temperature T the partition function sums the Boltzmann weight over all field configurations,

$$Z = \int D\theta \, \exp\left(- (K/2T) \int |\nabla\theta|^2 \right) . \quad (2.1)$$

For smooth (vortex-free) fields the exponent is quadratic, and (2.1) is a Gaussian integral over the orientation modes. On a lattice of N sites the Laplacian has eigenvalues λ_k, and the integral factorises mode by mode into

$$Z = \Pi_k \sqrt{ 2\pi T / (K \lambda_k) } , \quad (2.2)$$

the product running over the non-zero modes (the uniform mode is a free rotation and is omitted). This is the exact partition function of the Gaussian sector, and every equilibrium quantity of that sector follows from it by standard thermodynamic differentiation.

2a. Why the XY Functional Appears — the Correspondence Derived

The identification with the XY model should be earned, not asserted, since an outside reader will reasonably ask why exactly the rope orientation field carries the XY equilibrium functional. The derivation is short and rests only on structure already established in the microscopic-mechanics and homogenization papers.

- The local degree of freedom is an orientation ANGLE θ, valued on the circle S^1. This is the defining order-parameter space of an XY (planar-rotor) system — the same S^1 whose homotopy fixed the defect taxonomy.
- Nearest strands interact through an alignment energy that depends only on their relative angle and is minimised when aligned. The lowest even, periodic function of the angle

difference is $-J\cos(\Delta\theta)$; this is the XY nearest-neighbour coupling, and it is exactly the endpoint-locking energy the microscopic-mechanics paper derived ($J = T^2/\kappa$).

- In the small-relative-angle limit, $\cos(\Delta\theta) \approx 1 - \frac{1}{2}(\Delta\theta)^2$, so the coupling becomes $-J + (J/2)(\Delta\theta)^2$ up to a constant. Summing over neighbours and coarse-graining — the step the homogenization derivation made rigorous — turns $\Sigma(\Delta\theta)^2$ into the continuum $(K/2)\int|\nabla\theta|^2$ with $K = J/a$.
- Therefore the coarse-grained rope orientation field has the SAME equilibrium energy functional as the continuum XY model. The two share their Boltzmann weight, so their equilibrium statistical mechanics coincide.

This is an equivalence of EFFECTIVE EQUILIBRIUM theories: the coarse-grained field is governed by the XY functional. It is not the stronger claim that the microscopic rope medium is literally a lattice of XY spins — the rope medium has its own microscopic content (strands, tension, endpoint mechanics), and the XY functional is what that content coarse-grains TO in equilibrium. The correspondence is earned through the same coarse-graining the rest of the chain uses, which is why the statistical mechanics below is the rope medium's own, not an analogy borrowed from elsewhere.

> **The correspondence, precisely**
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> CLAIMED: the coarse-grained orientation field has the same equilibrium functional $(K/2)\int|\nabla\theta|^2$ as the 2D XY model, so their equilibrium statistical mechanics coincide (S^1 angle → $\cos(\Delta\theta)$ coupling → small-angle $(\nabla\theta)^2$ → continuum XY).
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> NOT CLAIMED: that the microscopic rope medium IS a lattice XY model. The equivalence is of effective equilibrium theories, established by coarse-graining.

3. Free Energy, Entropy, Energy, and Specific Heat

From the Gaussian partition function (2.2), the free energy per site is

$$F/N = -(T/2N)\, \Sigma_k \ln\left(2\pi T / (K \lambda_k) \right) , \tag{3.1}$$

which the computation evaluates and finds convergent as the lattice grows ($F/N \to -0.332$ in natural units at $T = K = 1$). The standard thermodynamic relations then give the rest of the sector:

3.1 Entropy and internal energy

The entropy is $S = -\partial F/\partial T$ and the internal energy $U = F + TS$. Computing S by numerical differentiation of (3.1) and forming U yields $U = 0.498$ per site at $T = 1$ — which matches the equipartition prediction of ½ T per quadratic mode to machine precision (0.498 vs 0.498). That the Gaussian sector obeys equipartition exactly is a direct check that the partition function (2.2) is correct.

3.2 Specific heat

The spin-wave specific heat is $C = \partial U/\partial T = \frac{1}{2}$ per mode, the two-dimensional analogue of the Dulong–Petit law: each quadratic orientation mode contributes ½ to the heat capacity. This is the smooth-sector contribution; the defect sector adds the singular behaviour near the transition (Section 6).

Gaussian sector — computed
Free energy $F/N = -(T/2N)\Sigma \ln(2\pi T/K\lambda_k)$, convergent ($\to -0.332$ at $T=K=1$).
Entropy $S=-\partial F/\partial T$; energy $U=F+TS=0.498$ = equipartition ½T per mode (exact check).
Specific heat $C=\partial U/\partial T$ = ½ per mode (2D Dulong–Petit).

4. The Defect Gas and the BKT Transition

The Gaussian sector alone has no phase transition — it describes quasi-ordered fluctuations at all temperatures. The transition comes from the defect (vortex) sector developed in the defect-theory paper: the medium supports vortices of energy $\pi K \ln(R/a)$ bound in neutral pairs at low temperature and liberated at high temperature. Treating the vortices as a gas of logarithmically interacting charges gives the Berezinskii–Kosterlitz–Thouless transition at

$$T_{BKT} = \pi K / 2 , \quad (4.1)$$

computed as 1.571 in natural units. The mechanism is an energy–entropy balance: a single free vortex costs energy $\pi K \ln(R/a)$ but carries entropy $\sim 2 \ln(R/a)$ from the $\ln(R/a)^2$ positions available to its core, so the free energy of an isolated vortex $F = (\pi K - 2T) \ln(R/a)$ changes sign exactly at $T = \pi K/2$. Below it, free vortices are forbidden and defects bind in pairs; above it, they proliferate.

Why this ties the sectors together. The transition temperature is set by the SAME stiffness K that governs electromagnetism (via the coarse-grained action) and the defect energetics (via the vortex energy). Completing thermodynamics therefore does not just fill one gap: it supplies the finite-temperature behaviour of every sector built on K. The defect-theory paper described the vortices; this paper puts them at finite temperature and derives when they unbind.

5. Correlations and the Universal Exponent

The signature of the low-temperature phase is quasi-long-range order: correlations decay not exponentially but as a power law,

$$\langle \cos(\theta(r) - \theta(0)) \rangle \sim r^{-\eta} , \quad \eta = T / 2\pi K , \quad (5.1)$$

with a temperature-dependent exponent — a rare and distinctive feature. At the transition the exponent reaches a universal value, computed as

$$\eta(T_{BKT}) = 1/4 , \quad (5.2)$$

exactly, independent of microscopic detail. Above T_{BKT} the decay crosses over to exponential, $\langle \cos\Delta\theta \rangle \sim \exp(-r/\xi)$, signalling the loss of order. The universal value $\eta = 1/4$ at the transition is one of the sharpest, most testable predictions in the whole continuum sector, and the computation returns it to nine decimal places.

6. Renormalization Flow and the Universal Jump

The defect gas is analysed by the Kosterlitz renormalization-group recursion for the stiffness K and the vortex fugacity y:

$$dK^{-1}/dl = 4\pi^3 y^2 , \quad dy/dl = (2 - \pi K)\, y , \qquad (6.1)$$

under coarse-graining scale l. The computation confirms the two phases as the two flow directions: for $\pi K > 2$ (below T_BKT) the fugacity flows to zero — vortices bound, order preserved — while for $\pi K < 2$ (above T_BKT) it grows — vortices unbind, order destroyed. The flow drives the stiffness to a renormalized value K_R that jumps discontinuously to zero at the transition, from the universal value

$$K\_R(T\_BKT) = 2\, T\_BKT / \pi , \qquad (6.2)$$

which the computation checks is self-consistently equal to K (both 1.000 in natural units) — the famous universal jump of the helicity modulus. Above the transition K_R = 0: the medium has no orientational stiffness and is disordered.

Defect sector — computed

BKT transition T_BKT = $\pi K/2$ = 1.571 (energy–entropy balance of a free vortex).

Correlation exponent $\eta = T/2\pi K$; η(T_BKT) = 1/4 exactly (universal).

KT RG flow (6.1) separates bound (low-T) from unbound (high-T); helicity jump K_R(T_BKT)=2T_BKT/π=K.

7. What Completing Thermodynamics Strengthens

Thermodynamics was worth completing not for its own sake but because several sectors depend on it. With the statistical mechanics in hand, each gains its finite-temperature footing:

Sector	What it gains from this paper
Defect theory	the finite-T fate of vortices: binding below T_BKT, unbinding above; the RG flow
Electromagnetism	the temperature dependence of the stiffness K that sets the EM coefficients
Condensed matter	the full BKT phase diagram, correlation exponent, and universal jump
Homogenization	the Boltzmann-weighted (not just minimised) continuum functional

The single functional $(K/2)\int|\nabla\theta|^2$ now carries a complete equilibrium thermodynamics, and the stiffness K that appears everywhere in the continuum chain is the same K whose renormalization drives the transition. The chain is thermodynamically closed.

8. The Thermodynamics Sector at a Glance

Quantity	Result (computed)
Partition function (Gaussian)	$Z = \Pi_k \sqrt{(2\pi T/K\lambda_k)}$

Free energy	$F/N = -(T/2N)\Sigma \ln(2\pi T/K\lambda_k)$
Entropy / energy	$S=-\partial F/\partial T$; $U=F+TS=0.498$ (equipartition)
Specific heat	$C = ½$ per mode
BKT transition	$T_{BKT} = \pi K/2 = 1.571$
Correlation exponent	$\eta = T/2\pi K$; $\eta(T_{BKT}) = 1/4$
Helicity jump	$K_R(T_{BKT}) = 2T_{BKT}/\pi = K$
RG flow	bound ($\pi K>2$) vs unbound ($\pi K<2$)

9. Scope and What Is Not Claimed

- The Gaussian sector's partition function, free energy, entropy, energy, and specific heat are derived and computed exactly. The defect-sector results (transition, exponent, jump, flow) are the established Berezinskii–Kosterlitz–Thouless results for the two-dimensional XY model, which the rope orientation field realises; we compute them on our functional and cite the XY correspondence rather than re-derive the full non-perturbative theory.
- The treatment is two-dimensional, matching the defect-theory paper; the finite-temperature physics of 3D defect lines and loops (from the 3D-topology paper) is a natural extension not developed here.
- This is classical equilibrium statistical mechanics. It does not touch the quantum boundary in either direction.

10. Status of Each Result

Result	Status
Gaussian partition function $Z = \Pi\sqrt{(2\pi T/K\lambda_k)}$	Derived — Gaussian integral; computed
Free energy, entropy, internal energy	Derived — from Z; equipartition check exact
Spin-wave specific heat $C = ½$ per mode	Derived — computed
BKT transition $T_{BKT} = \pi K/2$	Derived — energy–entropy balance; computed
Universal exponent $\eta(T_{BKT}) = 1/4$	Derived — computed to 9 digits
Helicity-modulus universal jump	Derived — computed, self-consistent
Kosterlitz RG flow, two phases	Derived — recursion integrated; phases separate
Full non-perturbative XY partition function	Modeled — XY correspondence, cited
Anything quantum	Out of scope — documented boundary, untouched

Conclusion

The EQUILIBRIUM statistical mechanics of the continuum orientation field is now complete, on the same functional the rest of the continuum chain uses and with every result computed. The Gaussian sector supplies the partition function as an exact Gaussian integral, and from it the free energy, entropy, internal energy — matching equipartition to machine precision — and the spin-wave specific heat. The defect sector supplies the finite-temperature physics: the Berezinskii–Kosterlitz–Thouless transition at $T_{BKT} = \pi K/2$ from the energy–entropy balance of a free vortex, the algebraic correlation exponent reaching the universal value $\eta = 1/4$ at the transition, the universal helicity-modulus jump $K_R(T_{BKT}) = 2T_{BKT}/\pi$ self-consistently equal to K, and the Kosterlitz renormalization flow that cleanly separates a low-temperature phase of bound defect pairs from a high-temperature phase of free defects. Because the transition is the defect-gas unbinding of the defect-theory paper, and the stiffness K is the same one that sets the electromagnetic coefficients and the vortex energetics, completing thermodynamics closes the continuum chain thermodynamically and strengthens the defect, electromagnetic, and condensed-matter sectors at once. The treatment stays within classical equilibrium statistical mechanics, and the one place it leans on established results — the full XY correspondence — is cited rather than overclaimed. What had been the biggest remaining hole in the foundations — the equilibrium thermodynamics of the continuum field — is now a computed, reproducible sector; nonequilibrium dynamics remains open and outside this paper's scope.

Registered Claims in This Paper

Each statement below is traceable to the machine-readable registry (claims.yaml), the corpus's single source of truth. Status labels: Derived (follows from stated mechanics), Modeled (reproduces data with declared inputs), EFT-constrained (bounded by effective-theory limits), Conjecture (proposed, not derived), Open (registered unsolved problem), Failed (falsified, kept as a finding). Where a claim is code-backed, the named benchmark reruns it.

- **THM-002 [Derived]:** Equilibrium Gaussian partition function Z=prod sqrt [benchmark: benchmarks/micromech/statistical_mechanics.py]
- **THM-003 [Derived]:** Spin-wave specific heat C = 1/2 per mode [benchmark: benchmarks/micromech/statistical_mechanics.py]
- **THM-004 [Derived]:** BKT transition T_BKT = pi K/2; universal correlation exponent eta [benchmark: benchmarks/micromech/statistical_mechanics.py]

- **THM-005 [Derived]:** Helicity-modulus universal jump K_R [benchmark: benchmarks/micromech/statistical_mechanics.py]

Programme credit: the core Rope Hypothesis is due to Bill Gaede; this work develops and formalises it. Computational collaboration and drafting by Claude (Anthropic). Correspondence to palmer100@gmail.com.